

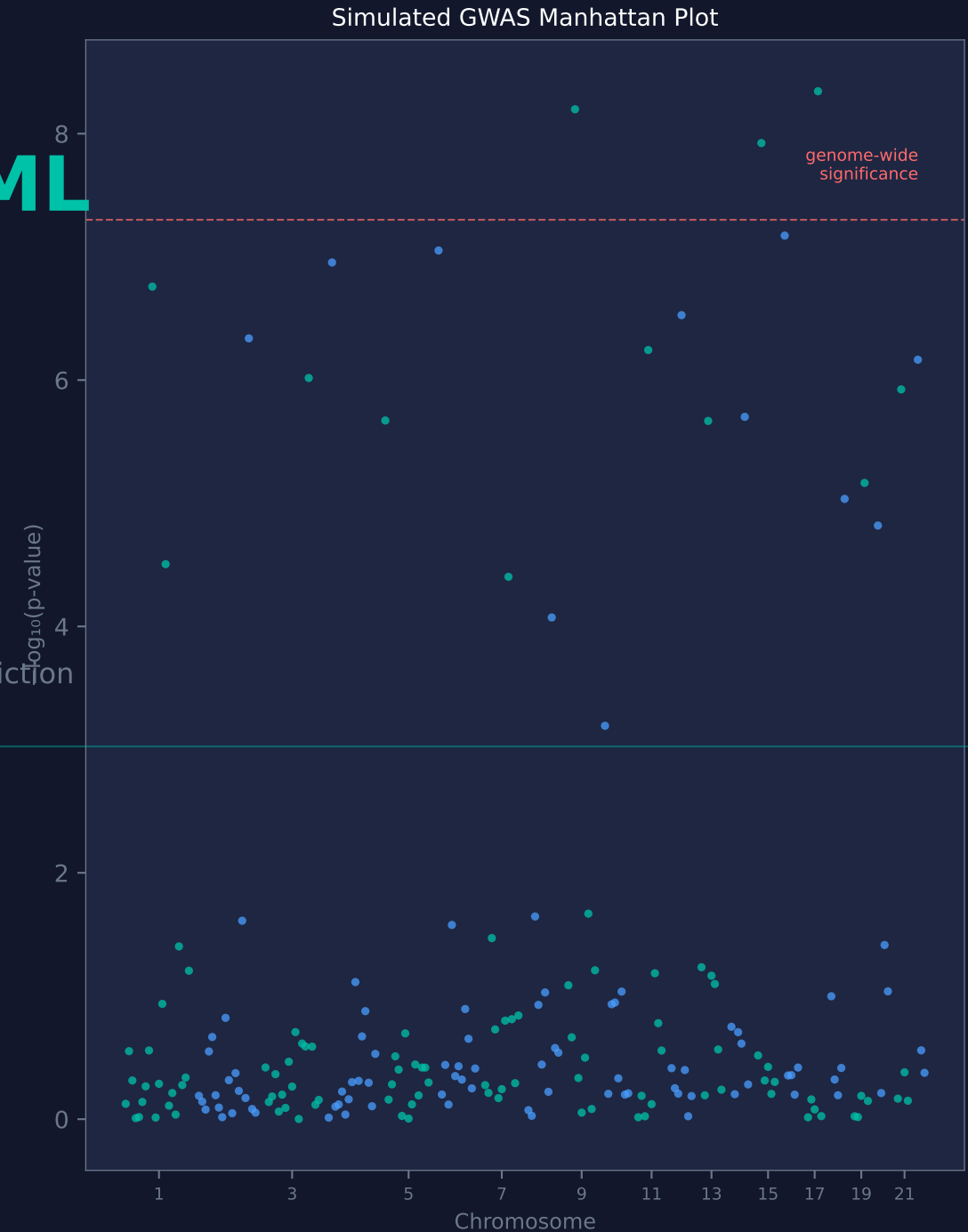
DEDA_MPC_GenomicML

Privacy-Preserving Machine Learning over Genomic Data

Multi-Party Computation · Secret Sharing · Disease Risk Prediction

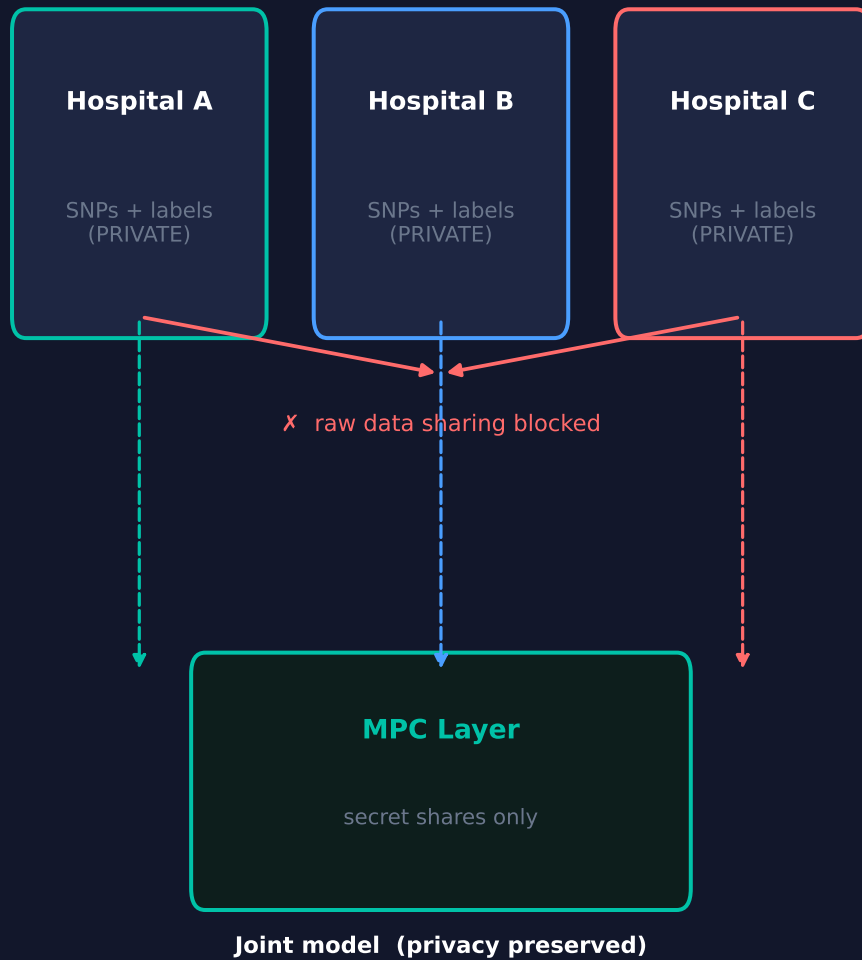
Pavel Shibaev

Digital Economy & Decision Analytics

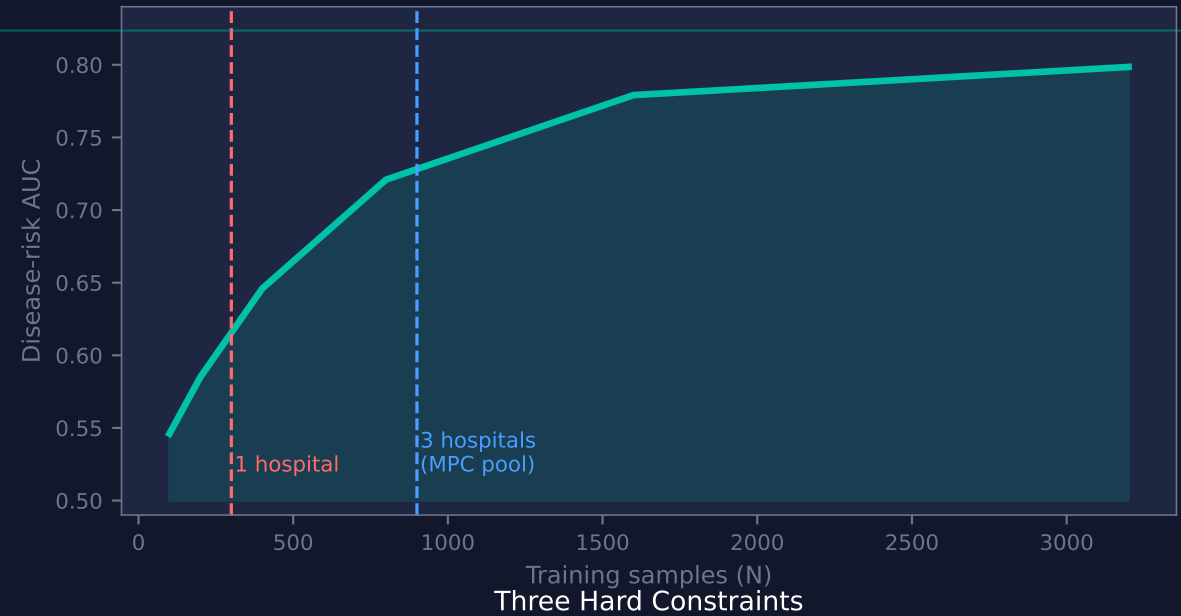


The Problem

Data-Silo Problem



Why Pooling Data Matters



Constraint

Consequence

Sensitivity

Genome = permanent ID; re-ID risk is real

Siloed data

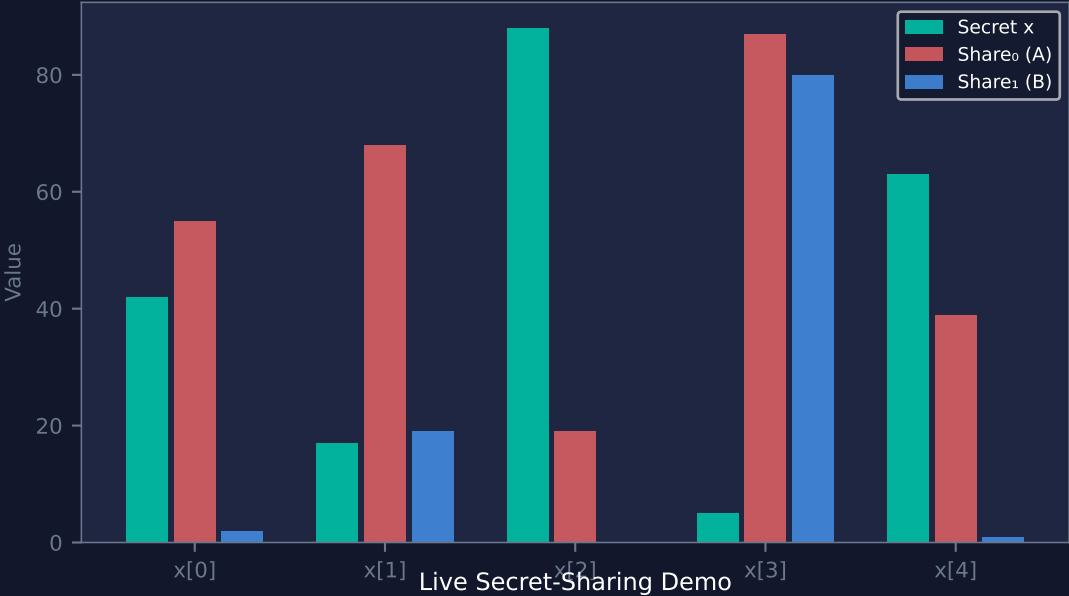
Each hospital too small to train reliably

Scale matters

3× cohort → significantly better AUC & PRS

Multi-Party Computation — How It Works

Secret vs. Shares (mod 100 for display)



Live Secret-Sharing Demo

x = [9 77 65 44]

y = [43 86 9 70]

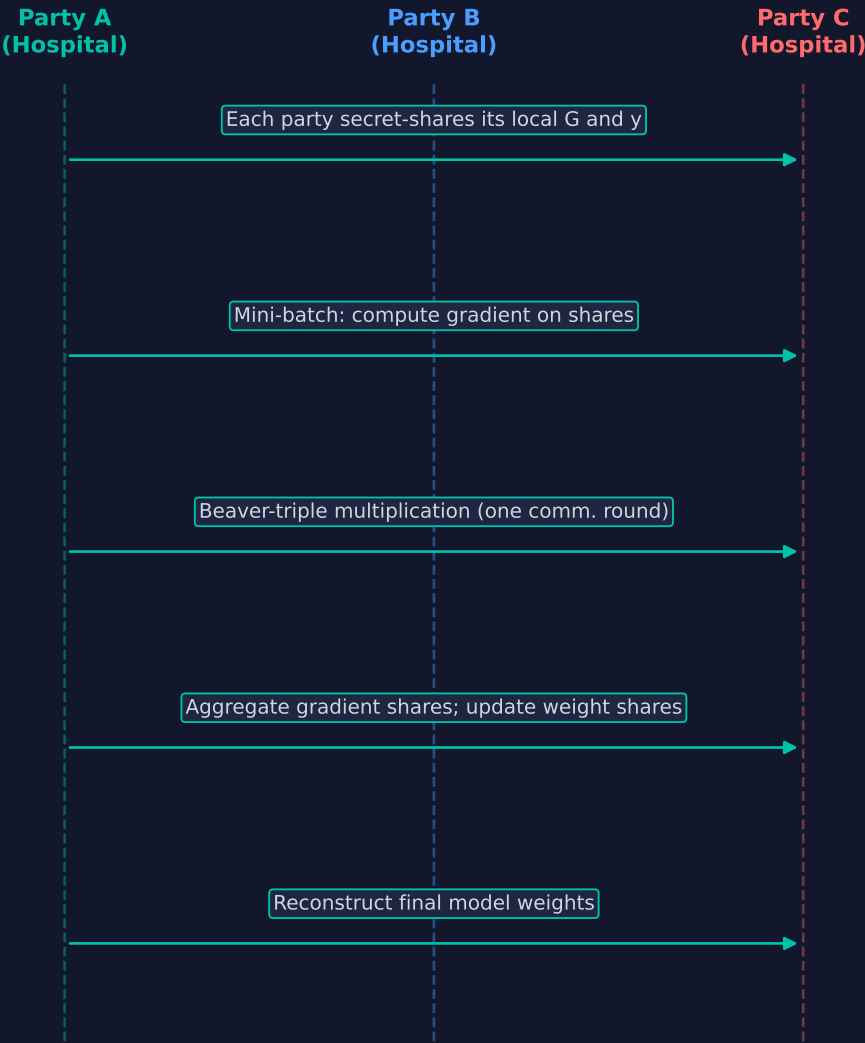
$x + y$ (plain) = [52 163 74 114]

MPC add = [52 163 74 114] ✓

$x \times y$ (plain) = [387 6622 585 3080]

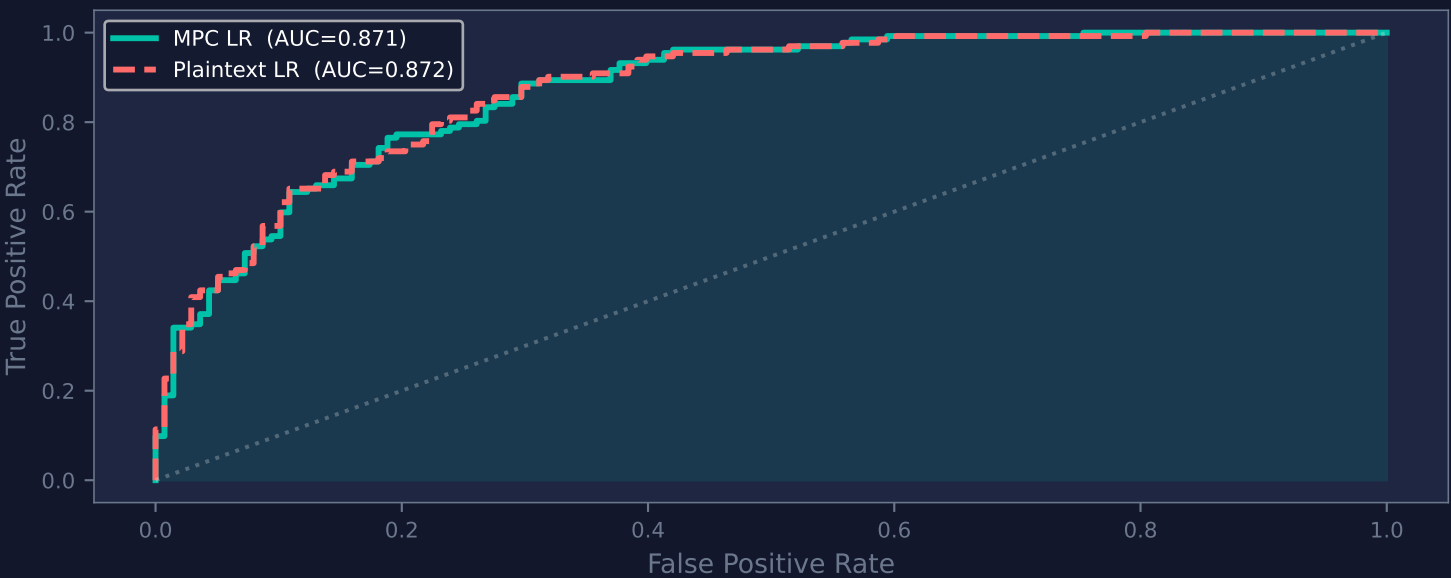
MPC mul = [17878841873010823 17502125873566018 10242558718145709 6852350672575715] ✓

3-Party MPC Protocol Sketch



Demo 1 — Secure Logistic Regression for Disease Risk

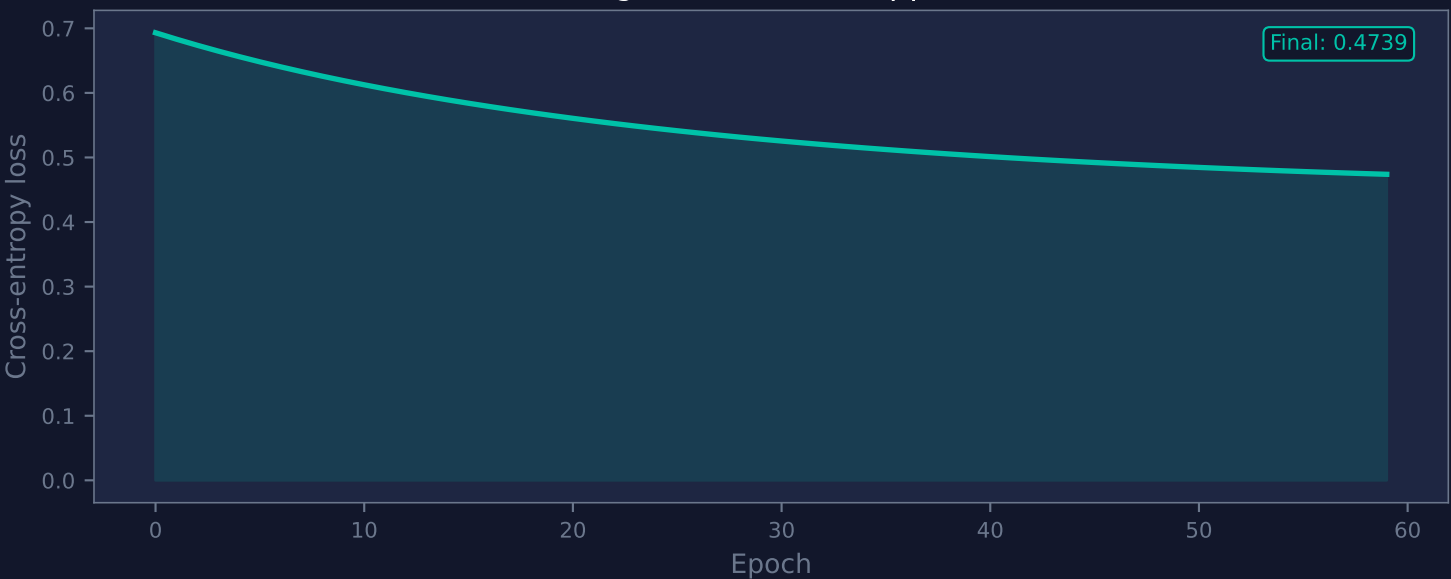
ROC — Disease Risk Prediction (50 SNPs)



AUC Comparison



MPC Training Loss (linear σ approximation)

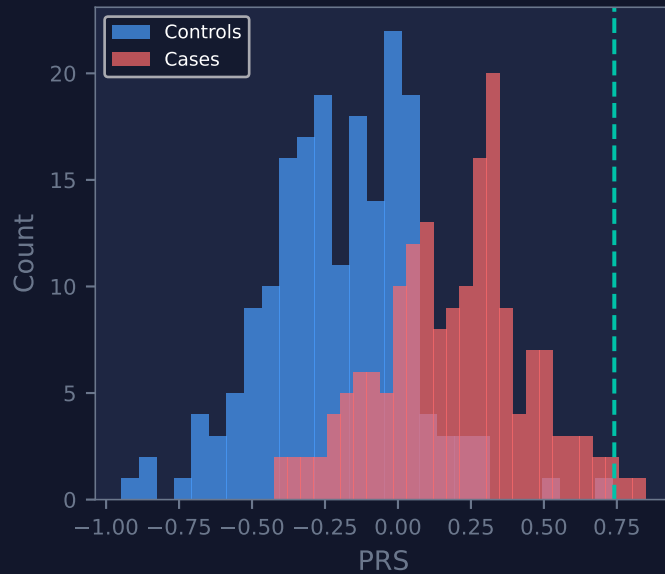


Run Configuration

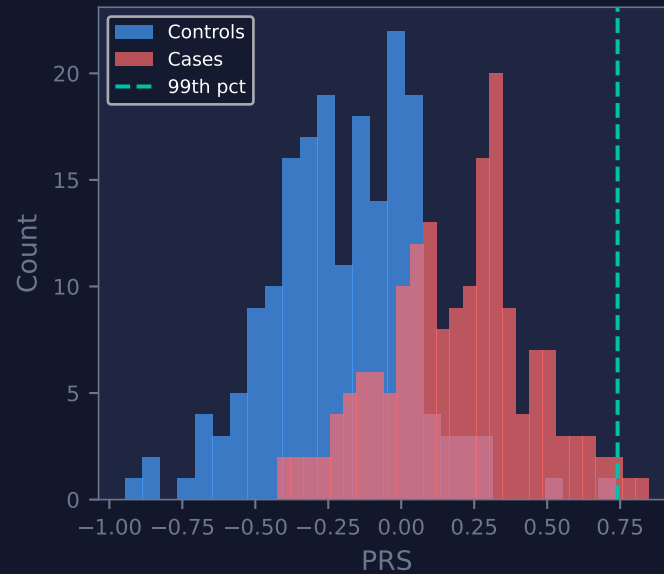
Samples	900 (3 parties $\times$ 300)
SNPs	50 (10 causal)
Heritability	$h^2 = 0.30$
Epochs	60
σ approx	$0.5 + 0.25t$
AUC gap	0.0017
Privacy cost	negligible

Demo 2 — Privacy-Preserving PRS & Final Takeaways

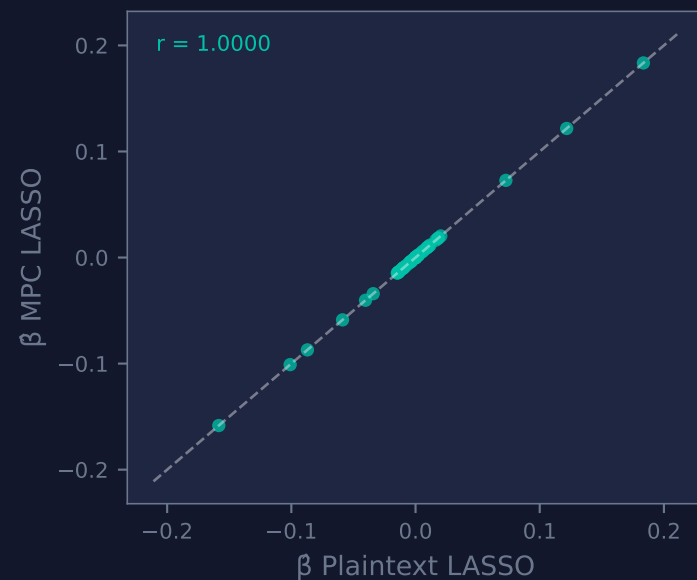
Plaintext PRS (AUC=0.868)



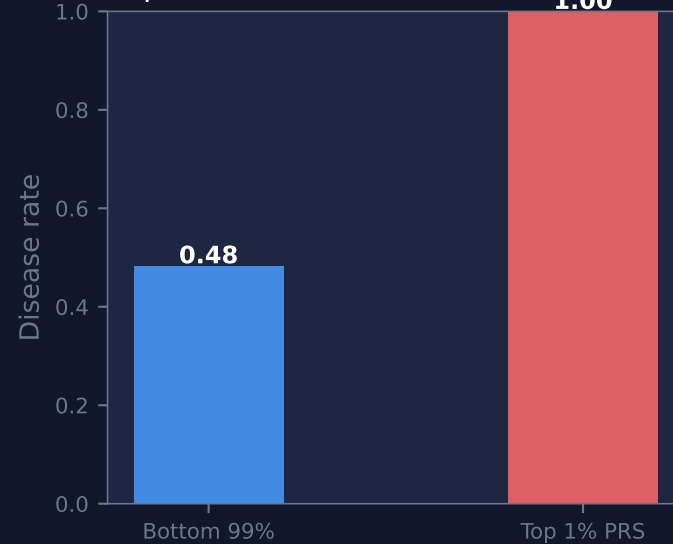
MPC PRS (AUC=0.868)



Effect Sizes: Plain vs. MPC



Top-1% PRS Enrichment ($2.1\times$ risk)



Key Takeaways

Clinical Impact

Multi-hospital models stratify patients before symptoms appear

Negligible Loss

< 0.002 AUC penalty for full cryptographic privacy

Regulatory Ready

GDPR / HIPAA: raw genomes never leave custodian infra

Open & Reproducible

[1] Mohassel & Zhang (2017) SecureML, IEEE S&P
Pure NumPy — no crypto lib required;
extend to MPC via SPDZ

[3] Privé et al. (2022) LDpred2, PLoS Genetics